

(PROJECT NAME)

Ten-Minute Presentation Script

Dam-break and landslide-dam inundation modelling by hydrodynamic simulation — automated setup, ensemble uncertainty, and outputs a district officer can act on.

TOTAL RUNTIME

10:00

PACE

~140 wpm

SLIDES

8

NUMBERS TO LAND

3

Every number in this version is one you can defend with a file in the repository. The numbers that were in the earlier draft but had never been measured — a 56-minute end-to-end time, a 17× speed-up, a 100-member ensemble, validation against observed satellite flood extents — have been replaced with measured equivalents. The claim is smaller. It is also unbreakable, which on a judging panel is worth more.

READING THE CUES

PAUSE — stop speaking for a full second, look up. **Bold** — land this word. **Coloured figures** — a measured number; say it precisely. Blue boxes are delivery notes, amber boxes are the honest caveat to have ready if a judge presses.

Good morning. We are Team Bit Loss, and we are solving Problem Statement 26161 — dam-break inundation modelling using hydrodynamic simulation. Our solution is called **(PROJECT NAME)**.

One line to set the stage: when a dam breaks, or when a landslide blocks a river and that blockage bursts, the villages downstream have somewhere between twenty minutes and two hours.

PAUSE

Today, getting a flood prediction ready takes longer than the flood takes to arrive — and most of that time is not the computer running. It is a specialist preparing the data by hand. **That gap is the entire problem.** Let me show you what we built to close it.

Right now, dam-break flood risk in India is handled with tools like Delft3D and HEC-RAS. These are excellent tools. They are also **the wrong shape for an emergency**. Here is why.

First — the setup is measured in weeks, not the run. Before you get a single result, an expert has to hand-prepare the terrain, build a computational mesh, and do the GIS work. That is the bottleneck, and it needs a specialist you may not have.

Second — they fail exactly when you need them. Most modern tooling is cloud-dependent. In a disaster, the network is the first thing to go down.

Third — landslide dams go completely unseen. A landslide blocks a river in a remote Himalayan valley. There is no sensor there. There is nobody to report it. The lake builds up quietly for days, and then it bursts. Chamoli in 2021 began like this.

Fourth — the terrain data goes stale. After a landslide the valley has physically changed, but the model is still using an old elevation map. So the answer it gives you is wrong — and it is **confidently** wrong, which is worse.

And fifth — you get one number, with no confidence attached. A single peak-flow figure. No range. No worst case. An officer has to order an evacuation based on a number nobody can put error bars on.

PAUSE

So: manual, fragile, blind, stale, and overconfident. That is what we set out to fix.

CHANGED FROM THE EARLIER DRAFT — AND WHY

The old opening line was "a single run takes two to six hours — water does not wait." Do not say it. Our own 300 m ensemble takes **5 hours**; a judge who knows these tools will ask, and the honest answer undercuts the slide. Lead with **setup weeks** instead — that is the gap we actually close, and it is a bigger one.

SLIDE 3 / 4

The Solution & What Makes It Different

2:45 – 5:15 · 2.5 min

This is (PROJECT NAME). Let me walk you through what it actually does, in order.

You give it four things.

Latitude and longitude, dam height, storage capacity, and breach type. That is the whole input. Nothing else.

PAUSE

Compare that to weeks of manual setup.

It fetches its own data.

Copernicus 30-metre elevation, GHSL population grids, Sentinel-1 radar, ESA land cover. All open, all free, no logins, no field survey. **The step that used to take an expert a week now takes the software a few minutes, by itself.**

It builds the breach on its own.

Instead of one assumed breach shape, we run a Monte Carlo ensemble across three published embankment breach regressions — Froehlich, MacDonald and Xu-Zhang — with Wahl's

uncertainty bands, plus Costa for natural landslide dams. We are not guessing. We are sampling the range of what the published physics says is possible.

Then it simulates in two parts.

Near the dam, where water is violent and three-dimensional, we hand off to a 3D SPH particle solver. Downstream, for tens of kilometres, we use a fast 2D shallow-water solver. This hybrid is the key engineering choice — full 3D everywhere would take days; 2D everywhere gets the breach physics wrong. We use each where it belongs.

SAY IT EXACTLY LIKE THIS

The handoff is **one-way**: shallow water feeds the particle solver, not the reverse. Never claim two-way coupling — it is not implemented, and it is the first thing a hydraulics academic on the panel will test you on. "One-way handoff, and that is a deliberate scope decision" is a strong answer. "Fully coupled" is a losing one.

We run an ensemble, then we check it.

Every run is an ensemble of members, and you get **p05**, **p50** and **p95** — a best case, a likely case and a worst case, with the member count printed next to them. Our demonstration runs use four to six members; the engine parallelises across workers, so the number is a budget decision, not a limit.

And we cross-check the solver itself against a **real Deltares kernel**. Same dam-break problem, run through our solver and through Delft3D FM, both scored against the exact analytical solution.

PAUSE

That is the part I want the judges to notice. **We are not asking you to trust our model. We are showing you our model scored against a published exact solution, next to the industry standard, on the same problem.**

And the output is something a responder can use.

Depth maps, velocity maps, and arrival-time maps — how many minutes until the water reaches your village. Exported as GeoTIFF, Shapefile and KML, so it drops straight into the GIS systems district authorities already use. And it works offline. On the right you can see the prototype running — 2D in Leaflet, 3D in Cesium. **This is built, not proposed.**

The comparison table underneath is the honest summary. Against Delft3D, HEC-RAS, DualSPHysics and the commercial tools — no existing system does automated setup, uncertainty ensembles, landslide-dam handling, terrain updating and satellite cross-checking **together**. Individually these exist. Together, in one pipeline a non-expert can run — that is what is new.

Briefly, how it is built. The numerical core is Python with NumPy, accelerated with Numba on the flux kernel — near-C speed, but readable and open. **No GPU requirement**, because a district office does not have one. GDAL handles the geospatial data. FastAPI serves the backend. The dashboard uses Leaflet for 2D and Cesium for the 3D view. Google Earth Engine gives us Sentinel-1 imagery for landslide detection and the observed-water overlay.

The important point about this stack: **every component is free and open-source**. There is no licence to buy. A state disaster authority can deploy this with zero procurement.

The solver uses HLLC shock-capturing with Audusse hydrostatic reconstruction — that is what makes it well-balanced: still water stays still, and a dam-break shock stays sharp instead of smearing out. Those are established, peer-reviewed methods. **We did not invent new physics. We engineered known-good physics into something fast and automatic.**

IF YOU ARE RUNNING OVER TIME

Cut from "The solver uses HLLC..." to the end of the slide. Keep the free-and-open-source line — it is the one a government evaluator remembers.

We have four real risks, and a concrete answer to each.

Risk one — full 3D simulation is computationally expensive. Our answer is the hybrid domain split: 3D only in the small violent zone near the breach, 2D for the long downstream reach.

Risk two — power and connectivity fail during a disaster. Our answer is offline-first architecture. Terrain is cached ahead of time, and the manual landslide-barrier path needs no network at all. Once your district is downloaded, (PROJECT NAME) runs with the network completely dead.

Risk three — terrain data goes stale after a landslide. Our answer is an observation-conditioned DEM update: we burn the detected barrier into the elevation model and write a new GeoTIFF carrying its own provenance, so the simulation runs on today's valley, not last year's.

Risk four — random sampling can hide the worst case. If you sample uncertain parameters independently you can miss the dangerous combination where several go bad at once. So we sample them jointly, which keeps the true worst case inside the ensemble.

None of these need new research. They are engineering decisions, and they are already implemented in the prototype.

HAVE THIS READY

The DEM update is **not photogrammetry** and every product we ship says so — it is Copernicus GLO-30 with a barrier burned in, every pixel outside the footprint bit-identical to source. We use Sentinel-1 GRD change detection, not InSAR, because Earth Engine does not carry SLC. Volunteering that limitation reads as rigour.

SLIDE 7

Impact and Benefits

7:45 – 9:30 · 1.75 min

Now the part that matters — what changes on the ground.

The headline is that the human bottleneck disappears. Current practice needs an expert working for days to weeks to prepare the terrain, the mesh and the GIS before anything runs at all. (PROJECT NAME) does that setup itself, unattended, in minutes.

PAUSE

And then it solves. Here are our measured wall-clock times, on a sixteen-core machine with no GPU:

46 min

DECISION-GRADE RUN

28 × 26 km domain, 200 m grid, 30 h of flood simulated, 6-member ensemble.

64 min

WIDE-DOMAIN SCREEN

240 × 188 km domain, 500 m grid, 24 h simulated, 4 members.

5 h

FINE-GRID CONFIRMATION

Same domain at 300 m. We report the grid with the time, every time.

So the honest claim is this: **the setup goes from weeks to minutes, and the solve is under an hour for a decision-grade run.** Sixteen hours of expert preparation plus a run means you model the flood after it happened. Automated setup plus a forty-six-minute solve means you model it while there is still time to move people.

IF A JUDGE ASKS "WHY IS 300 M SLOWER THAN 200 M?"

Because the domain is nine times larger, not because the grid is finer — the 300 m run covers 240 × 188 km against 28 × 26 km. Say that plainly and give the cell counts if pushed. Never quote a time without its grid, its domain and its member count; that pairing is what makes the number checkable.

The four impacts.

Lives saved, because earlier warnings mean villages get evacuated in time. **Hidden dangers found**, because we spot landslide dams forming where nobody is watching. **Lower disaster**

losses, because damage and risk maps let authorities pre-position before the water arrives. And **better preparedness** — dam safety can be tested on any ordinary Tuesday, not only in the middle of an emergency.

Where we are today.

Done: the 2D solver is gated on analytical tests and cross-checked against a real Delft3D FM kernel; both dam break and river blockage are handled; the dashboard exports GeoTIFF, Shapefile and KML; **600** tests pass. **In progress:** real-terrain benchmarks — Malpasset and Chamoli — near real-time analysis through Earth Engine, and damage estimation, whose cost coefficients are currently labelled unvetted in the output rather than dressed up as published. **Next:** generalising to any river and any dataset, national-scale data volumes, and Emergency Action Plan map packs in the CWC format — which means the output slots directly into the government's existing process instead of asking anyone to change how they work.

SLIDE 8

Research and References

9:30 – 10:00 · 30 sec

Everything here rests on published work — CWC and CDSO guidelines for dam flood mapping, the Chamoli reconstructions in *Science* and *Natural Hazards*, Toro and Audusse for the numerical solver, Froehlich, Xu-Zhang and Wahl for the breach parameters, and Maranzoni's 2023 review, which recommends exactly the coupled 2D–3D decomposition we implemented.

All our data sources are open and free, and we deliberately excluded datasets with restrictive licences so the outputs can be legally redistributed by a government agency. And every number on this deck is reproducible — the validation script and the run commands are on the slide.

PAUSE

To close: existing tools tell you what the flood **was**. (PROJECT NAME) tells you what the flood **will be** — with a confidence range, without a specialist, on a laptop, with the network down. Thank you.

SLIDE	TOPIC	START	LENGTH
1-2	Title	0:00	0:45
3	Problem	0:45	2:00
3 / 4	Solution + Innovation	2:45	2:30
5	Technical Approach	5:15	1:15
6	Feasibility & Risks	6:30	1:15
7	Impact & Benefits	7:45	1:45
8	References + Close	9:30	0:30

IF YOU ARE RUNNING OVER

Cut the HLLC and hydrostatic-reconstruction detail on slide 5, and shorten slide 8 to one sentence.

Never cut the setup-weeks-to-minutes line or the 46-minute run on slide 7 — together they are your strongest claim, and unlike the old 17× figure, both survive being checked.

Three numbers you must land

NUMBER	SAY IT AS	BACKED BY
4 inputs	"Latitude, longitude, dam height, storage, breach type — that is the whole input."	Run request schema
46 minutes	"Forty-six minutes, 200 metre grid, six-member ensemble, sixteen cores, no GPU."	Measured run e2e09ea3, 2,740 s
p05 / p50 / p95	"Every output carries a best, likely and worst case, with the member count next to it."	Ensemble statistics in the run summary

Four questions to be ready for

IF THEY ASK	ANSWER
"Have you validated against a real flood?"	Not yet on real terrain — we are scored against the Ritter exact solution and cross-checked engine-to-engine with Delft3D FM. Malpasset and Chamoli are the next benchmarks, and they are set up. Do not claim satellite flood-extent validation; that comparison has not been run.
"Is the SPH real, or a picture?"	The near-field solver is real PySPH and runs on the coupled path. The SPH curve in the older comparison figure was synthesised for layout — that figure is not shown as a result, and we can say so.
"How many ensemble members?"	Four to six in our demonstration runs, parallelised across workers. The limit is RAM per worker, roughly 400 MB, not the algorithm.
"What is your worst known bug?"	A 24-hour run held water that should have drained — we traced it to an unbreached dam ridge in the DEM, resampling pits, and a domain with no downstream to drain into. After the fix, 96.4% of released volume exits and severe hazard reaches zero at 9.4 hours. Volunteer this. Finding and fixing your own defect is the single most credible thing you can say to a technical panel.

(PROJECT NAME) · SIH 2026 · PS 26161 · Team Bit Loss — every figure in this script is traceable to a run log or a test in the repository.